INFORMATION COMMUNICATION TECHNOLOGY

NETWORK DESIGN PRACTICAL 3,TERM 3

MODULE 3

Time allowed: 4 hours
Tools and materials:

Cisco Packet Tracer,

- ✓ A digital multimeter,
- ✓ Straight through cable
- ✓ a cable tester (optional).
- ✓ Cross over cable

Instructions to the Candidate

1. Work individually and read all tasks before you begin.
2. Complete the tasks in the order given.
3. Record all readings and observations in your answer sheet.
4. Raise your hand to have your work checked by the examiner when you see the words **"Call the examiner"**.
5. Save your Packet Tracer file as YourName_VLAN_Practical.pkt.

Addressing Table

Device	VLAN	Switch Port	IP Address	Subnet Mask
PC1	10 (ACCOUNTS)	Switch1 Fa0/1	192.168.10.10	255.255.255.0
PC2	10 (ACCOUNTS)	Switch2 Fa0/1	192.168.10.11	255.255.255.0
PC3	20 (HR)	Switch1 Fa0/3	192.168.20.10	255.255.255.0
PC4	20 (HR)	Switch2 Fa0/3	192.168.20.11	255.255.255.0

Trunk link: **Switch1 Fa0/24 ↔ Switch2 Fa0/24**

Task 1: Test Ethernet Cable Continuity Using a Multimeter

Objective: Confirm that each cable is fit for use before it is connected to the network.

Instructions:

1. Take one straight-through cable. Hold both RJ-45 connectors with the clips facing down and the contacts facing you, so that **Pin 1 is on the left**.
2. Switch on the multimeter and turn the dial to the **continuity mode**, marked with the buzzer or diode symbol.
3. Touch the two probes together. The meter should beep. This confirms that the multimeter is working.
4. Place one probe on **Pin 1 of Connector A** and the other on **Pin 1 of Connector B**. Note whether the meter beeps (continuity) or stays silent (open circuit).
5. Repeat for pins 2 to 8, always testing the same pin number at both ends.
6. Then test one pair of different pins on the same connector (for example Pin 1 and Pin 2) to check for a short circuit. The meter should not beep.
7. Record your results in the table below.

Pin at Connector A	Pin at Connector B	Beep? (Yes/No)	Result (Good/Faulty)
1	1		
2	2		
3	3		
4	4		
5	5		
6	6		
7	7		
8	8		

8. Test the **crossover cable** the same way. This time, expected continuity is Pin 1 → Pin 3, Pin 2 → Pin 6, Pin 3 → Pin 1, Pin 6 → Pin 2, and Pins 4, 5, 7 and 8 straight across. Record your results.

9. State whether each cable is **good** or **faulty**, and name the cable type.

Call the examiner to verify your readings.

Task 2: Build the Topology and Configure IP Addresses

Instructions:

1. Open Cisco Packet Tracer and add **2 Cisco 2960 switches** and **4 PCs**.
2. Use **Copper Straight-Through** cables to connect each PC to its switch port as shown in the Addressing Table.
3. Use a **Copper Cross-Over** cable to connect Switch1 Fa0/24 to Switch2 Fa0/24.
4. Rename the switches **Switch1** and **Switch2**.
5. Assign the IP address and subnet mask to each PC as shown in the Addressing Table (Desktop → IP Configuration).
6. Wait for all link lights to turn green, then take a screenshot of the topology.

Task 3: Create VLANs

Instructions:

1. Open the CLI of **Switch1** and enter privileged configuration mode.
2. Create VLAN 10 and name it **ACCOUNTS**.
3. Create VLAN 20 and name it **HR**.
4. Exit to privileged EXEC mode.
5. Use the appropriate command to display a brief summary of the VLANs and confirm that both VLANs appear.
6. Repeat steps 1 to 5 on **Switch2**.
7. Copy the output of the verification command into your answer sheet.

Task 4: Assign Ports to VLANs

Instructions:

1. On **Switch1**, configure interface Fa0/1 as an access port and assign it to VLAN 10.
2. On **Switch1**, configure interface Fa0/3 as an access port and assign it to VLAN 20.
3. On **Switch2**, configure interface Fa0/1 as an access port and assign it to VLAN 10.
4. On **Switch2**, configure interface Fa0/3 as an access port and assign it to VLAN 20.
5. Verify the port assignments on both switches using the VLAN summary command.
6. In your answer sheet, write the purpose of the commands switchport mode access and switchport access vlan 10.

Call the examiner to check the VLAN port assignments.

Task 5: Test Communication Before Trunking

Instructions:

1. Before configuring the trunk, make sure the Fa0/24 ports are **not** in trunk mode.
2. From **PC1**, ping PC2 (192.168.10.11). Record the result.
3. From **PC3**, ping PC4 (192.168.20.11). Record the result.
4. From **PC1**, ping PC3 (192.168.20.10). Record the result.
5. In your answer sheet, explain the results you obtained in steps 2 to 4.

Task 6: Configure and Verify Trunking

Instructions:

1. On **Switch1**, enter interface configuration mode for Fa0/24 and set the port to **trunk** mode.
2. Repeat the same configuration on **Switch2**.
3. Verify the trunk using the command that displays trunk interfaces. Copy the output into your answer sheet and note the mode, encapsulation and status of Fa0/24.
4. Repeat the following tests and record the results:

- From **PC1**, ping PC2 (192.168.10.11).
- From **PC3**, ping PC4 (192.168.20.11).
- From **PC1**, ping PC3 (192.168.20.10).
- From **PC2**, ping PC4 (192.168.20.11).

5. Compare these results with those in Task 5 and answer the following in your answer sheet:
   - (a) What changed after the trunk was configured, and why?
   - (b) Why do PCs in different VLANs still fail to communicate, even with the trunk in place?
   - (c) What device would be needed to allow communication between VLAN 10 and VLAN 20?

Call the examiner to observe the ping tests.

Task 7: Documentation and Saving

Instructions:

1. Save the running configuration on both switches using the appropriate command.
2. Save the Packet Tracer file with the required file name.
3. Submit your answer sheet containing the multimeter readings, command outputs, screenshots and written explanations.